

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

EE/EX-7103-GS

B.E. VII Semester

Examination, December 2020

Grading System (GS)

SCADA Systems and Applications

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1.
 - a) Draw the block diagram of PLC and explain each block.
 - b) Describe the ladder diagram application and method to draw it.
2. Draw and explain block diagram of Programmable logic controller.
3.
 - a) Explain in detail remote terminal unit.
 - b) Discuss about Intelligent Electronic Devices.
4.
 - a) What is the State Estimation Process of power system? Explain.
 - b) What are open systems interconnection protocols?
5.
 - a) Describe Energy Management System. What are the basic functions of Energy Control Centre?
 - b) What do you know about the three operating state of power system? Discuss each of them with the help of suitable diagram.

EE/EX-7103-GS

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[2]

6.
 - a) What is Automatic substation control system.
 - b) What are secure state and insecure state of operation.
7.
 - a) Mention any one case study for the application of SCADA of the following:
 - i) Gas industry
 - ii) Oil industry
 - iii) Transmission and distribution of electrical energy
 - b) Explain on-line and off line functions in an EMS.
8. Write short note on any two.
 - i) PLC-SCADA interfacing
 - ii) RTU
 - iii) Fibre-optics communication
